AIRY

FLIGHT ANALYSIS YAW SNAPS AND VIBRATION

Oct 20 test flights, five phone videos against their card logs. Prepared Oct 21, 2026.

BOTTOM LINE

Vibration does not cause the yaw snaps. Every snap is the estimator (EKF2) forcing its heading onto a wrong compass reading, and the video shows the aircraft turning smoothly through them. The vibration is a separate **17 Hz roll oscillation** in the control loop.

FLIGHTS ANALYSED

Videos matched to logs by motor sound and recording order. Figures cover powered flight only, takeoff to kill.

VIDEO	LOG	POWERED FLIGHT (S)	HEADING RESETS (°)	MAX COMPASS ERROR (°)	GYRO VIBRATION 10 TO 40 HZ (°/S)	TURNED IN FLIGHT (°)
IMG_7520	log_409	41	−20	71	6.0	−348
IMG_7524	log_413	23	+48	175	5.4	−485
IMG_7525	log_414*	40	+83, −155	177	8.0	−744
IMG_7529	log_423	72	+54	168	2.6	−367
IMG_7530	log_424	25	+97, −149	170	3.2	−542

* Most likely log_414. log_417 flew the same plan 14 minutes later and cannot be ruled out. Red marks the two large snaps.

HEADING SNAPS COMPASS RESETS

EKF2 resets its heading to the compass in two situations, and both appear in these flights:

- **Height trigger, every flight.** 6 to 8.5 s after takeoff, the first climb past 1.5 m triggers a one-off realignment to the compass.
- **Rejection timeout, logs 414 and 424.** After 3 to 4 s of rejected compass readings, EKF2 gives up and takes the compass value: a jump of about −150°.

The compass error most likely comes from the motors' current and the indoor field (0.37 to 0.44 G at rest, varying by spot), not vibration: its error correlates with vibration at only −0.1 to 0.3.



IMG_7525 at the −155° snap: 0.5 s before, at the snap, and 0.5 s and 1.0 s after (left to right). The aircraft (top left) turns smoothly; nothing jumps.

VIBRATION 17 HZ ROLL OSCILLATION

A narrow line at 17.0 Hz on the roll axis dominates the gyro, 800 to 2,700 times the background. The six rear motors carry it; the front three do not.

- **Not fan vibration.** It holds at 17.0 Hz while throttle and fan speed (486 to 590 Hz blade tone) change, and the fan noise does not wobble at 17 Hz.
- **Not linked to the compass.** Compass errors do not rise or fall with it.
- **Accelerometer within range.** 0.7 to 1.7 m/s^2 in the 10 to 60 Hz band, no clipping. Short bursts come at takeoff and just before the kill, on contact with the mat or floor.

On Oct 19 the same oscillation sat at 26 to 30 Hz with unchanged roll gains. Since then the yaw model, yaw gains and hover thrust changed, and the ATLAS module was enabled on the board. It is either the loop itself or a frame mode the loop excites; these logs cannot tell which.

YAW DRIFT, NOT SATURATION

The Oct 19 saturation is gone: the tilted rear fans are never pinned, and PX4 gets 89 to 98% of the torque it asks for. The aircraft still turns the same way every flight, 350° to 740° in total. Yaw-rate P is now 0.07 (was 0.2), and the heading it holds keeps moving with the compass.

NEXT STEPS

1. Set **EKF2_MAG_TYPE** to 6 (compass at startup only) or 5 (never) for indoor flights. This removes both kinds of snap.
2. Fly one hop with roll-rate P at 0.3 instead of 0.45. If the 17 Hz line moves or shrinks, it is the loop.
3. Log the raw high-rate IMU on that hop (SDLOG_PROFILE with high-rate sensors).
4. Film from a tripod a few metres back, keeping the whole flight in frame.